

UPS Mini System - Project Danger Log

Database Risks

1. Data Integrity Issues

- *Risk*: Foreign key constraints might fail if records are deleted improperly
- *Mitigation*: Implement proper ON DELETE actions and transaction management

2. Database Connection Pool Exhaustion

- *Risk*: Under high load, connection pool may become exhausted
- *Mitigation*: Monitor connection usage, implement connection timeouts

Communication Risks

3. World Simulator Connection Failures

- *Risk*: Connection to World Simulator may drop unexpectedly
- *Mitigation*: Implement robust reconnection logic with exponential backoff

4. Protocol Buffer Parsing Errors

- *Risk*: Malformed messages from World Simulator could cause parsing errors
- *Mitigation*: Add exception handling around all Protocol Buffer operations

5. Amazon/UPS Communication Failures

- *Risk*: API communication between Amazon and UPS may fail
- *Mitigation*: Implement message queuing with retries and proper error handling

Real-time Map Feature Risks

6. Canvas Performance Issues

- *Risk*: Map rendering may slow down with large number of elements
- *Mitigation*: Implement spatial indexing and only render visible elements

7. Browser Compatibility

- *Risk*: Canvas implementation may vary across browsers
- *Mitigation*: Test on multiple browsers, use feature detection

8. User Location Data Privacy

- *Risk*: Displaying exact package locations may raise privacy concerns
- *Mitigation*: Limit location precision for non-owners of packages

Notification System Risks

9. Email Delivery Failures

- *Risk*: Email notifications might not be delivered or marked as spam
- *Mitigation*: Implement proper SPF/DKIM setup, monitor delivery rates

10. Notification Timing Accuracy

- *Risk*: Estimated delivery times may be inaccurate, leading to poorly timed notifications
- *Mitigation*: Implement adaptive timing based on historical delivery data

11. User Preference Data Loss

- *Risk*: User notification preferences might be lost during updates
- *Mitigation*: Implement versioning for preference settings and backup procedures

General Application Risks

12. Authentication Vulnerabilities

- *Risk*: Authentication system may be susceptible to attacks
- *Mitigation*: Use Django's built-in authentication with proper password policies

13. CSRF/XSS Vulnerabilities

- *Risk*: Forms might be vulnerable to CSRF or XSS attacks
- *Mitigation*: Ensure Django's CSRF protection is enabled, validate all input

14. Performance Under Load

- *Risk*: System may slow down with many concurrent users
- *Mitigation*: Implement caching, optimize database queries, use asynchronous processing

15. Environment Configuration Issues

- *Risk*: Different environments (dev/test/prod) might behave differently
- *Mitigation*: Use environment variables, maintain consistent configurations

Integration Risks

16. Dependency Version Conflicts

- *Risk*: Different library versions may cause compatibility issues
- *Mitigation*: Use virtual environments and pin dependency versions

17. Docker Container Communication

- *Risk*: Inter-container networking issues might prevent proper communication
- *Mitigation*: Define proper network settings, use health checks

18. Django Template Rendering Errors

- *Risk*: Template errors might only be discovered at runtime
- *Mitigation*: Implement comprehensive template testing

Project Management Risks

19. Scope Creep

- *Risk*: Additional features beyond requirements may delay completion
- *Mitigation*: Maintain strict prioritization, implement core functionality first

20. Integration Group Communication Failures

- *Risk*: Poor communication between Amazon and UPS implementation teams
- *Mitigation*: Regular sync meetings, documented API contracts, shared testing

This danger log identifies the major risks associated with the UPS Mini System implementation, with particular focus on the real-time delivery visualization and notification system features that serve as our product differentiation. Regular review of these risks throughout development will help ensure successful implementation.